40.1: Animal form and function are correlated at all levels of organization

Over the course of its life, an Emperor penguin faces the same fundamental challenges as any other animal, whether hydra, hawk, or human. All animals must obtain nutrients and oxygen, fight off infection, and survive to produce offspring. Given that all animal species share these and other basic requirements, why does their form, including **anatomy**—biological structure—vary so widely? The answer lies in natural selection and adaptation. Natural selection favors those variations in a population that increase relative fitness (see Concept 23.4). The evolutionary adaptations that enable survival vary among environments and species but frequently result in a close match of form to function, as illustrated for the Emperor penguins in Figure 40.1. Because structure and function are correlated, examining anatomy often provides clues to **physiology**—biological function.

An animal's size and shape are fundamental aspects of form that significantly affect the way the animal interacts with its environment. Although we may refer to size and shape as elements of a "body plan" or "design," this does not imply a process of conscious invention. The body plan of an animal is the result of a pattern of development programmed by the genome, itself the product of millions of years of evolution.

Evolution of Animal Size and Shape

EVOLUTION Many different body plans have arisen during the course of evolution, but these variations fall within certain bounds. Physical laws that govern strength, diffusion, movement, and heat exchange limit the range of animal forms.

As an example of how physical laws constrain evolution, let's consider how some properties of water limit the possible shapes for animals that are fast swimmers. Water is about 1,000 times denser than air and also far more viscous. Therefore, any bump on an animal's body surface that causes drag impedes a swimmer more than it would a runner or flyer. Tuna and other fast ray-finned fishes can swim at speeds up to 80 km/hr (50 miles/hour). Sharks, penguins, dolphins, and seals are also relatively fast swimmers. As illustrated by the three examples in **Figure 40.2**, these animals all have a shape that is fusiform, meaning tapered on both ends. The similar streamlined shape found in these speedy vertebrates is an example of convergent evolution (see Concept 22.3). Natural selection often results in similar adaptations when diverse organisms face the same environmental challenge, such as overcoming drag during swimming.

Figure 40.2

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Convergent evolution in fast swimmers.
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A seal, penguin, and tuna all have similar body shapes when swimming.

Physical laws also influence animal body plans with regard to maximum size. As body dimensions increase, thicker skeletons are required to maintain adequate support. This limitation affects internal skeletons, such as those of vertebrates, as well as external skeletons, such as those of insects and other arthropods. In addition, as bodies increase in size, the muscles required for locomotion must represent an ever-larger fraction of the total body mass. At some point, mobility becomes limited. By considering the fraction of body mass in leg muscles and the effective force such muscles generate, scientists can estimate maximum speed for a wide range of body plans. In the case of the 6-meter-tall dinosaur *Tyrannosaurus rex*, there is controversy, with some scientists calculating a top running speed as fast as that of an Olympic sprinter—30 km/hr (19 miles/hour), but others inferring that *T. rex* was at best a fast walker.

Exchange with the Environment

Animals must exchange nutrients, waste products, and gases with their environment, and this requirement imposes an additional limitation on body plans. Exchange occurs as substances dissolved in an aqueous solution move across the plasma membrane of each cell. A single-celled organism, such as the amoeba in **Figure 40.3a**, has a sufficient membrane surface area in contact with its environment to carry out all necessary exchange. In contrast, an animal is composed of many cells, each with its own plasma membrane across which exchange must occur. The rate of exchange is proportional to the membrane surface area involved in exchange, whereas the amount of material that must be exchanged is proportional to the total body volume. A multicellular organization therefore works only if every cell has access to a suitable aqueous environment, either inside or outside the animal's body.

Figure 40.3

Direct exchange with the environment.

Two organisms and their environments are compared.

Part a. An amoeba, a single-celled organism, exchanges nutrients, wastes and gasses with the environment across its membrane. Scale: 0.1 millimeters.

Part b. Hydra, an animal with two layers of cells, is a cylindrical-shaped animal with tentacles on top and a mouth at the base of the tentacles. Hydra cross section taken from the middle of the animal is a circle two cell layers thick with a gastrovascular cavity in the center. Exchange occurs at the outer surface and inside the gastrovascular cavity that is also bathed in water. Scale: 1 millimeter.

Many animals with a simple internal organization have body plans that enable direct exchange between almost all their cells and the external environment. For example, a pond-dwelling hydra has a saclike body plan and a body wall only two cell layers thick **(Figure 40.3b)**. Because its gastrovascular cavity opens to the external environment, both the outer and inner layers of cells are constantly bathed by pond water. Another common body plan that maximizes exposure to the surrounding medium is a flat shape. Consider, for instance, a parasitic tapeworm, which can reach several meters in length (see Figure 33.11). A thin, flat shape places most cells of the worm in direct contact with its particular environment—the nutrient-rich intestinal fluid of a vertebrate host.

Our bodies and those of most other animals are composed of compact masses of cells, with an internal organization much more complex than that of a hydra or a tapeworm. For such a body plan, increasing the number of cells decreases the ratio of outer surface area to total volume. As an extreme comparison, the ratio of outer surface area to volume for a whale is hundreds of thousands of times smaller than that for a water flea. Nevertheless, every cell in the whale has access to oxygen, nutrients, and other resources. How is this accomplished?

In whales and most other animals, the evolutionary adaptations that enable sufficient exchange with the environment are specialized surfaces that are extensively branched or folded **(Figure 40.4)**. In almost all cases, these exchange surfaces lie within the body, an arrangement that protects their delicate tissues from abrasion or dehydration and allows for streamlined body contours. The branching or folding greatly increases surface area (see Figure 33.8). In humans, for example, the exchange surfaces for digestion, respiration, and circulation each have an area more than 25 times larger than that of the skin.

Figure 40.4

Internal exchange surfaces of complex animals.

Exchange surfaces of the circulatory, respiratory, excretory and digestive systems of complex animals. In all cases, extensive branching or folding increases the surface area for exchange.

Most animals have surfaces that are specialized for exchanging chemicals with the surroundings. These exchange surfaces are usually internal but are connected to the environment via openings on the body surface (the mouth, for example). The exchange surfaces are finely branched or folded, giving them a very large area. The digestive, respiratory, and excretory systems all have such exchange surfaces. Chemicals exchanged across these surfaces are transported throughout the body via the circulatory system.

Exchange surfaces of the circulatory, respiratory, excretory and digestive systems of complex animals. In all cases, extensive branching or folding increases the surface area for exchange. In the digestive system, food is taken in through the mouth and travels along the heavily folded digestive tract. The small intestine is lined with finger-like projections (shown in SEM at scale 100 micrometers) that expand the surface area for

nutrient absorption. Nutrients pass into the circulatory system from the digestive system, and unabsorbed matter (feces) pass out of the animal body through the anus. In the respiratory system, oxygen is taken in from the environment. A microscopic view of the lung (SEM, scale 250 micrometers) reveals that it is much more sponge-like than balloon like with an extensively branched, expansive wet surface for gas exchange with the environment. Carbon dioxide also passes out of the respiratory system and is expelled from the animal body. In the circulatory system, carbon dioxide passes from the blood to the lungs (from where it will be expelled) and oxygen passes from the lungs to the blood. As oxygen travels in the blood in the vessels of the circulatory system, exchange occurs with the body cells across the interstitial fluid (cells uptake oxygen and give off CO_2). The blood then travels back to the lungs to give off CO_2 and again pick up oxygen. Exchange also occurs between the circulatory system and the excretory system; within the kidney, blood is filtered across the surface of long, narrow blood vessels packed into ball-shaped structures. Microscopic view shows heavily folded (brain-like in appearance) balls clustered among branches of thinner strands (SEM, scale 50 micrometers). Metabolic waste products (nitrogenous waste) are expelled from the animal body.

VISUAL SKILLS *Using this diagram, explain how exchange carried out by animals can be described as both internal and external.*

*For suggested answer, see*Appendix A*.*

Internal body fluids link exchange surfaces to body cells. The spaces between cells are filled with fluid, known in many animals as **interstitial fluid** (from the Latin for “stand between”). Complex body plans also include a circulatory fluid, such as blood. Exchange between the interstitial fluid and the circulatory fluid enables cells throughout the body to obtain nutrients and get rid of wastes (see Figure 40.4).

Complex body plans offer numerous benefits. For example, an external skeleton can protect against predators, and sensory organs can provide detailed information on the animal’s surroundings. Internal digestive organs can break down food gradually, controlling the release of stored energy. In addition, specialized filtration systems can adjust the composition of the internal fluid that bathes the animal’s body cells. In this way, an animal can maintain a relatively stable internal environment despite the fact that it is living in a changeable external environment. A complex body plan is especially advantageous for animals living on land, where the external environment may be highly variable.

Hierarchical Organization of Body Plans

Cells form a working animal body through their *emergent properties*, which arise from successive levels of structural and functional organization (see Concept 1.1). Cells are organized into **tissues**, groups of cells with a similar appearance and a common function. Different types of tissues are further organized into functional units called **organs**. (The simplest animals, such as sponges, lack organs or even true tissues.)

Groups of organs that work together, providing an additional level of organization and coordination, make up an **organ system(Table 40.1)**. Thus, for example, the skin is an organ of the integumentary system, which protects against infection and helps regulate body temperature.

Table 40.1

Organ Systems in Mammals

A table titled Table 40.1 Organ system in mammals illustrates organ systems, main components, and main functions.

Table 40.1 Organ systems in Mammals
Organ System
Digestive
Circulatory
Respiratory
Immune and lymphatic
Excretory
Endocrine
Reproductive
Nervous
Integumentary
Skeletal
Muscular

Many organs have more than one physiological role. If the roles are distinct enough, we consider the organ to belong to more than one organ system. The pancreas, for instance, produces enzymes critical to the function of the digestive system but also regulates the level of sugar in the blood as a vital part of the endocrine system.

Just as viewing the body's organization from the "bottom up" (from cells to organ systems) reveals emergent properties, a "top-down" view of the hierarchy reveals the multilayered basis of specialization. Organ systems include specialized organs made up of specialized tissues and cells. Consider the human digestive

system: Each organ has specific roles. In the case of the stomach, one role is to initiate protein breakdown. This process requires a churning motion powered by stomach muscles, as well as digestive juices secreted by the stomach lining. Producing digestive juices, in turn, requires highly specialized cell types: One cell type secretes a protein-digesting enzyme, a second generates concentrated hydrochloric acid, and a third produces mucus, which protects the stomach lining.

The specialized and complex organ systems of animals are built from a limited set of cell and tissue types. For example, lungs and blood vessels have different functions but are lined by tissues that are of the same basic type and therefore share many properties.

There are four main types of animal tissues: epithelial, connective, muscle, and nervous. **Figure 40.5** explores the structure and function of each type. In later chapters, we'll discuss how these tissue types contribute to the functions of particular organ systems.

Figure 40.5 Interactive Visual Activity

Exploring Structure and Function in Animal Tissues

Interact with this figure in the Figure 40.5 Interactive Visual Activity: Exploring Structure and Function in Animal Tissues.

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Figure 40.5

Exploring Structure and Function in Animal Tissues

The figure is of a wolf and describes five types of epithelial tissue: stratified squamous, cuboidal, simple columnar, simple squamous, and pseudostratified columnar; also describes the polarity of epithelia.

Occurring as sheets of cells, epithelial tissues, or epithelia (singular, epithelium), cover the outside of the body and line organs and cavities within the body. Because epithelial cells are closely packed, often with tight junctions, they function as a barrier against mechanical injury, pathogens, and fluid loss. Epithelia also form active interfaces with the environment. For example, the epithelium that lines the nasal passages is crucial for olfaction, the sense of smell. Note how different cell shapes and arrangements correlate with distinct functions.

Stratified squamous epithelium: A stratified squamous epithelium is multilayered and regenerates rapidly. New cells formed by division near the basal surface (see micrograph below) push outward, replacing cells that are sloughed off. This epithelium is commonly found on surfaces subject to abrasion, such as the outer

skin and the linings of the mouth, anus, and vagina. Apical surface is top layer; basal surface is bottom cell layer.

Cuboidal epithelium: A cuboidal epithelium, with dice-shaped cells specialized for secretion, makes up the epithelium of kidney tubules and many glands, including the thyroid gland and salivary glands.

Simple columnar epithelium: The large, brick-shaped cells of simple columnar epithelia are often found where secretion or active absorption is important. For example, a simple columnar epithelium lines the intestines, secreting digestive juices and absorbing nutrients.

Simple squamous epithelium: The single layer of plate like cells that form a simple squamous epithelium functions in the exchange of material by diffusion. This type of epithelium, which is thin and leaky, lines blood vessels and the air sacs of the lungs, where diffusion of nutrients and gases is critical.

Pseudostratified columnar epithelium: A pseudostratified epithelium consists of a single layer of cells varying in height and the position of their nuclei. In many vertebrates, a pseudostratified epithelium of ciliated cells forms a mucous membrane that lines portions of the respiratory tract. The beating cilia sweep the film of mucus along the surface.

Polarity of epithelia: All epithelia are polarized, meaning that they have two different sides. The apical surface faces the lumen (cavity) or outside of the organ and is therefore exposed to fluid or air. Specialized projections often cover this surface. For example, the apical surface of the epithelium lining the small intestine is covered with microvilli, projections that increase the surface area available for absorbing nutrients. The opposite side of each epithelium is the basal surface. Image scale: 10 micrometers.

The figure is of a wolf and describes six types of connective tissue: loose connective, fibrous connective, bone, blood, adipose, and cartilage.

Connective tissue, consisting of a sparse population of cells scattered through an extracellular matrix, holds many tissues and organs together and in place. The matrix generally consists of a web of fibers embedded in a liquid, jellylike, or solid foundation. Within the matrix are numerous cells called fibroblasts, which secrete fiber proteins, and macrophages, which engulf foreign particles and any cell debris by phagocytosis. Connective tissue fibers are of three kinds: Collagenous fibers provide strength and flexibility, reticular fibers join connective tissue to adjacent tissues, and elastic fibers make tissues elastic. If you pinch a fold of tissue on the back of your hand, the collagenous and reticular fibers prevent the skin from being pulled far from the bone, whereas the elastic fibers restore the skin to its original shape when you release your grip. Different mixtures of fibers and foundation form the major types of connective tissue described below.

• Loose connective tissue: The most widespread connective tissue in the vertebrate body is loose connective tissue, which binds epithelia to underlying tissues and holds organs in place. Loose connective tissue gets its

name from the loose weave of its fibers, which include all three types. It is found in the skin and throughout the body. Image shows connective tissue with collagenous and elastic fibers. Image scale: 120 micrometers.

• Fibrous connective tissue: Fibrous connective tissue is dense with collagenous fibers. It is found in tendons, which attach muscles to bones, and in ligaments, which connect bones at joints. Image shows multiple nuclei interspersed in layers of fibers. Scale: 30 micrometers.

• Bone: The skeleton of most vertebrates is made of bone, a mineralized connective tissue. Bone-forming cells called osteoblasts deposit a matrix of collagen. Calcium, magnesium, and phosphate ions combine into a hard mineral within the matrix. The microscopic structure of hard mammalian bone consists of repeating units called osteons. Each osteon has concentric layers of the mineralized matrix, which are deposited around a central canal containing blood vessels and nerves. Image shows repeating osteons with layers around central canals. Scale: 700 micrometers.

• Blood: Blood has a liquid extracellular matrix called plasma, which consists of water, salts, and dissolved proteins. Suspended in plasma are erythrocytes (red blood cells), leukocytes (white blood cells), and cell fragments called platelets. Red cells carry oxygen, white cells function in defense, and platelets aid in blood clotting. Image shows red blood cells interspersed with plasma and white blood cells. Scale: 55 micrometers.

• Adipose tissue: loose connective tissue that stores fat in adipose cells distributed throughout its matrix. Adipose tissue pads and insulates the body and stores fuel as fat molecules. Each adipose cell contains a large fat droplet that swells when fat is stored and shrinks when the body uses that fat as fuel. Image shows fat droplets. Scale: 150 micrometers.

• Cartilage: Cartilage contains collagenous fibers embedded in a rubbery protein-carbohydrate complex called chondroitin sulfate. Cells called chondrocytes secrete the collagen and chondroitin sulfate, which together make cartilage a strong yet flexible support material. The skeletons of many vertebrate embryos contain cartilage that is replaced by bone as the embryo matures. Cartilage remains in some locations, such as the disks that act as cushions between vertebrae. Image shows chondrocytes interspersed with chondroitin sulfate. Scale: 100 micrometers.

The figure is of a wolf and describes the three types of muscle tissue: skeletal, smooth, and cardiac muscle. It also describes nervous tissue: neurons and glia.

The tissue responsible for nearly all types of body movement is muscle tissue. All muscle cells consist of filaments containing the proteins actin and myosin, which together enable muscles to contract. There are three types of muscle tissue in the vertebrate body: skeletal, smooth, and cardiac.

Skeletal muscle: Attached to bones by tendons, skeletal muscle, or striated muscle, is responsible for voluntary movements. Skeletal muscle consists of bundles of long cells that are called muscle fibers. During development, skeletal muscle fibers form byte fusion of many cells, resulting in multiple nuclei in each

muscle fiber. The arrangement of contractile units, or sarcomeres, along the fibers gives the cells a striped (striated) appearance. In adult mammals, building muscle increases the size but not the number of muscle fibers. Image shows nuclei interspersed between muscle fibers with repeated sarcomeres (appear as stripes). Scale: 100 micrometers.

Smooth muscle: Smooth muscle, which lacks striations, is found in the walls of the digestive tract, urinary bladder, arteries, and other internal organs. The cells are spindle shaped. Smooth muscles are responsible for involuntary body activities, such as churning of the stomach and constriction of arteries. Image shows nuclei interspersed with muscle fibers; no striations. Scale: 25 micrometers.

Cardiac muscle: Cardiac muscle forms the contractile wall of the heart. It is striated like skeletal muscle and has similar contractile properties. Unlike skeletal muscle, however, cardiac muscle has branched fibers that interconnect via intercalated disks, which relay signals from cell to cell and help synchronize heart contraction. Image shows cell nuclei and striations visible. Intercalated discs appear as lines connecting fibers. Scale: 25 micrometers.

Neurons: Neurons are the basic units of the nervous system. A neuron receives nerve impulses from other neurons via its cell body and multiple extensions called dendrites. Neurons transmit impulses to neurons, muscles, or other cells via extensions called axons, which are often bundled together into nerves. Image shows that neuron has long axon (stem-like in appearance) topped with a round cell body and with star-like extensions called dendrites. Scale: 40 micrometers.

Glia: The various types of glia help nourish, insulate, and replenish neurons, and in some cases, modulate neuron function. Image shows network of neuron axons interspersed with glia and blood vessels. Scale: 15 micrometers.

Animation

Overview of Animal Tissues

image

image

Animation

Epithelial Tissue

image

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image

Animation

Connective Tissue

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image

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Animation

Muscle Tissue

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Animation

Nervous Tissue

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Coordination and Control

For an animal's tissues and organ systems to function effectively, they must act in concert with one another. For example, when the wolf shown in Figure 40.5 hunts, blood flow is regulated to bring adequate nutrients and gases to its leg muscles, which in turn are activated by the brain in response to cues detected by the nose. What signals coordinate activity? How do the signals move within the body?

Animals have two major systems for coordinating and controlling responses to stimuli: the endocrine and nervous systems **(Figure 40.6)**. In the **endocrine system**, signaling molecules released into the bloodstream by endocrine cells are carried to all locations in the body. In the **nervous system**, neurons transmit signals along dedicated routes connecting specific locations in the body. In each system, the type of pathway used is the same regardless of whether the signal's ultimate target is at the other end of the body or just a few cell diameters away.

Figure 40.6

Signaling in the endocrine and nervous systems.

Signaling in the endocrine and nervous systems.

Part a. Signaling by hormones, which induce a response only in cells with specific receptors. Endocrine system: A stimulus induces an endocrine cell to secrete hormone signals. Hormone travels everywhere in the body via the bloodstream. Response to the hormone signal is limited to cells that have the receptor for the signal, as diagrammed by hormones leaving the blood vessel and only binding to specific cells with the correct receptor on their surface.

Part b. Signaling by neurons, which induce a response only in cells connected by specialized junctions. Nervous system: a stimulus is transmitted to a neuron, with a star-like cell body containing the nucleus and long axon tail, along which the nerve impulse signal travels to a specific location. A response is limited to cells that connect by special junctions to an axon that transmits an impulse, as diagrammed here by axons connecting to the outer surface of only some body cells.

VISUAL SKILLS*After comparing the two diagrams, explain why a particular nerve impulse signal has only one physical pathway but a particular hormone molecule can have multiple physical pathways.*

*For suggested answer, see*Appendix A*.*

The signaling molecules that are broadcast throughout the body by the endocrine system are called **hormones**. It takes seconds for hormones to be released into the bloodstream and carried throughout the body. The effects are often long-lasting, however, because hormones can remain in the bloodstream for minutes or even hours.

Different hormones cause distinct effects, and only cells that have receptors for a particular hormone respond (Figure 40.6a). Depending on which cells have receptors for that hormone, the hormone may have an effect in just a single location or in sites throughout the body. For example, thyroid-stimulating hormone (TSH) acts solely on cells in the thyroid gland. They in turn release thyroid hormone, which acts on nearly every body tissue to increase oxygen consumption and heat production.

In the nervous system, signals called nerve impulses travel to specific target cells along communication lines consisting mainly of axons (Figure 40.6b). Transmission in the nervous system is extremely fast; nerve impulses take only a fraction of a second to reach the target and last only a fraction of a second.

Nerve impulses can act on other neurons, on muscle cells, and on cells and glands that produce secretions. Unlike the endocrine system, the nervous system conveys information by the *pathway* the signal takes. For example, a person can distinguish different musical notes because each note's frequency activates neurons in the ear that connect to slightly different locations in the brain.

Communication in the nervous system usually involves more than one type of signal. Nerve impulses travel along axons, sometimes over long distances, as changes in voltage. In contrast, passing information from one neuron to another often involves very short-range chemical signals.

Because the two major communication systems of the body differ in signal type, transmission, speed, and duration, it is not surprising that they are adapted to different functions. The endocrine system is especially well adapted for coordinating gradual changes that affect the entire body, such as growth, development, reproduction, metabolic processes, and digestion. The nervous system is well suited for directing immediate and rapid responses to the environment, such as reflexes and other rapid movements. Nevertheless, the two systems often work in close coordination. Both help maintain a stable internal environment, our next topic of discussion.